MA385: Tutorial Sheet 4

These exercises are for Tutorial 4 (Week 8). You do not have to submit solutions to these questions. Similar questions may appear on the class test or final exam.

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Q1. For the following functions show that they satisfy a Lipschitz condition on the corresponding domain, and give an upper-bound for L:

(i) $f(t,y) = 2yt^{-4}$ for $t \in [1,\infty)$,

(ii) $f(t,y) = 1 + t\sin(ty)$ for $0 \leqslant t \leqslant 2$.

Q2. Suppose we use Euler's method to find an approximation for $y(2)$, where y solves

$$y(1) = 1, \qquad y' = (t-1)\sin(y).$$

(i) Give an upper bound for the global error taking $n = 4$ (i.e., $h = 1/4$).

(ii) What n should you take to ensure that the global error is no more that 10^{-3}?

Q3. (Based on a question from the 25/26 exam). Consider the general two-stage Runge-Kutta (RK2) method:

$$y_0 = y(t_0), \qquad y_{i+1} = y_i + h(ak_1 + bk_2) \quad \text{for } i = 0,1,2,\ldots,n-1, \tag{1}$$

where $k_1 = f(t_i, y_i)$, and $k_2 = f(t_i + \alpha h, y_i + \beta h k_1)$. The parameters a, b, α and β are chosen to give a specific method.

(a) Suppose y solves the initial value problem

$$y(1) = 1, \qquad y'(t) = 2t \quad \text{for} \quad t > 1.$$

Explain why the RK2 method should compute the exact solution. Use this fact to give an expression for α in terms of b.

(b) Suppose that we attempt to solve $y'(t) = \lambda y(t)$, with $y(0) = 1$, using a RK2 method. Use the fact that the RK2 solution should agree with the Taylor series for $y(t_{i+1})$ about t_i, up to terms of order h^2, to give an expression for β in terms of b.

(c) Take $a = 3/4$. Determine the values of b, α and β in (1) that give a RK2 method.

Q4. Here is the tableau for a three stage Runge-Kutta method:

$$\begin{array}{c|ccc} \alpha_1 & & & \\ \alpha_2 & \beta_{21} & & \\ \alpha_3 & \beta_{31} & \beta_{32} & \\ \hline & b_1 & b_2 & b_3 \end{array} = \begin{array}{c|ccc} 0 & & & \\ \alpha_2 & 1/2 & & \\ 1 & \beta_{31} & 2 & \\ \hline & 1/6 & b_2 & 1/6 \end{array}$$

(i) Use that the method is consistent to determine b_2.

(ii) The method is exact when used to compute the solution to

$$y(0) = 0, \quad y'(t) = 2t, \ t > 0.$$

Use this to determine α_2.

(iii) The method should agree with an appropriate Taylor series for the solution to $y'(t) = \lambda y(t)$, up to terms that are $\mathcal{O}(h^3)$. Use this to determine β_{31}.